Original MadGraph5_aMC@NLO diagrams

Charged current | double Higgs

Process: e- mu+ -> ve vm~ h h QCD=0

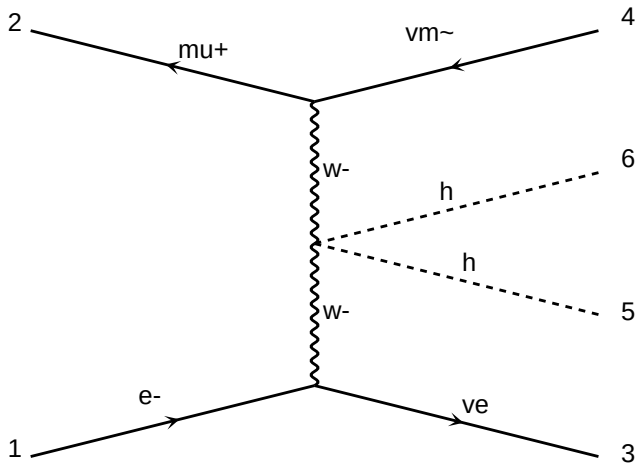

diagram 1 QCD=0, QED=4

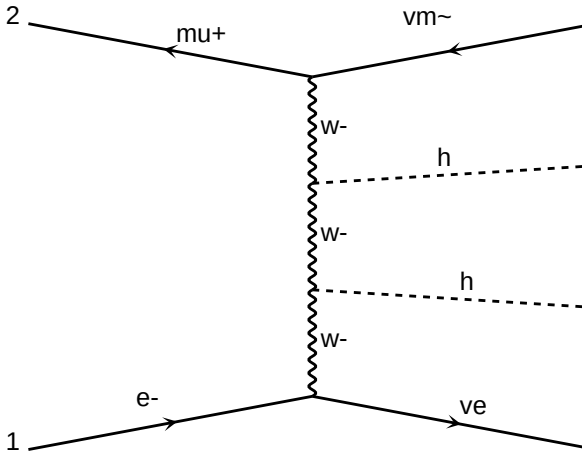

diagram 2 QCD=0, QED=4

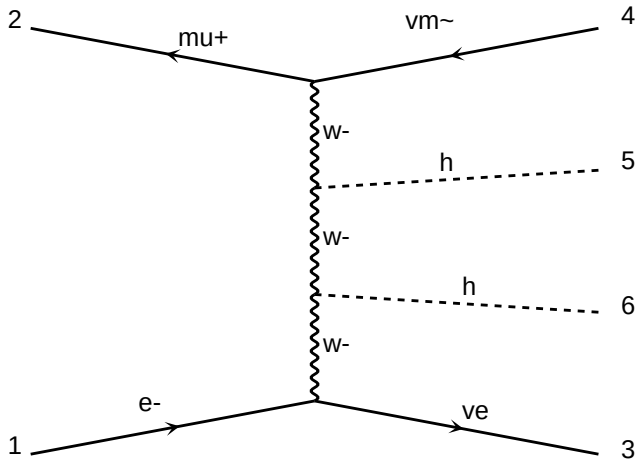

diagram 3 QCD=0, QED=4

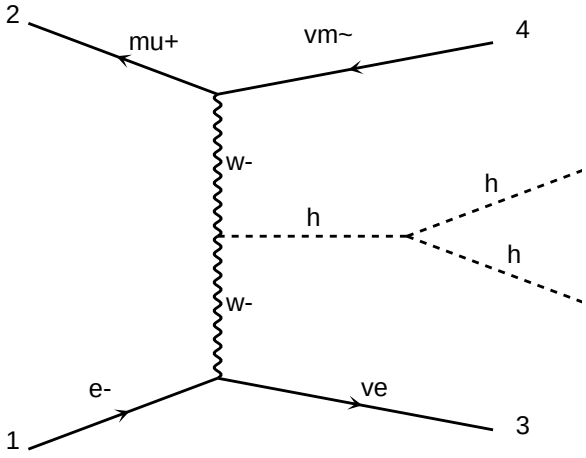

diagram 4 QCD=0, QED=4

What the numbered diagrams contain:

1 - HHWW contact interaction (two Higgs bosons at one vertex)

2, 3 - separate Higgs emission through two HWW couplings

4 - virtual Higgs H* -> HH (trilinear self-coupling)

Exact original MG5 diagram numbering retained; all four diagrams interfere.

Original MadGraph5_aMC@NLO diagrams

Neutral current | double Higgs

Process: e- mu+ -> e- mu+ h h QCD=0

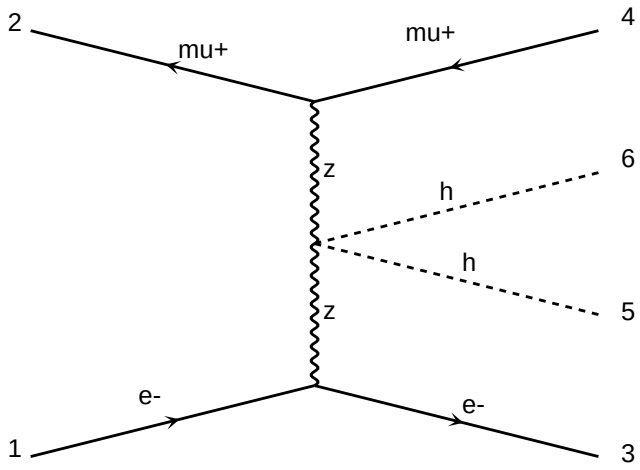

diagram 1 QCD=0, QED=4

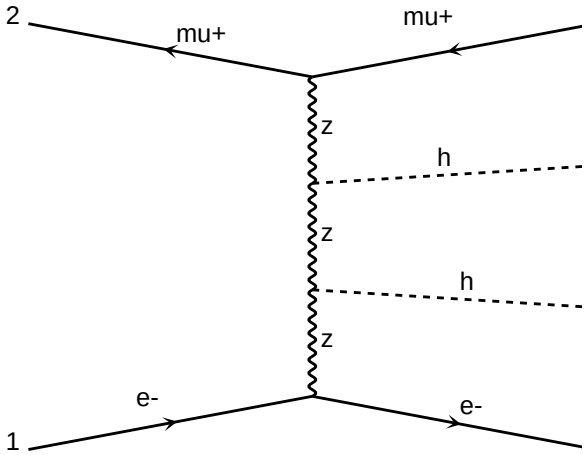

diagram 2 QCD=0, QED=4

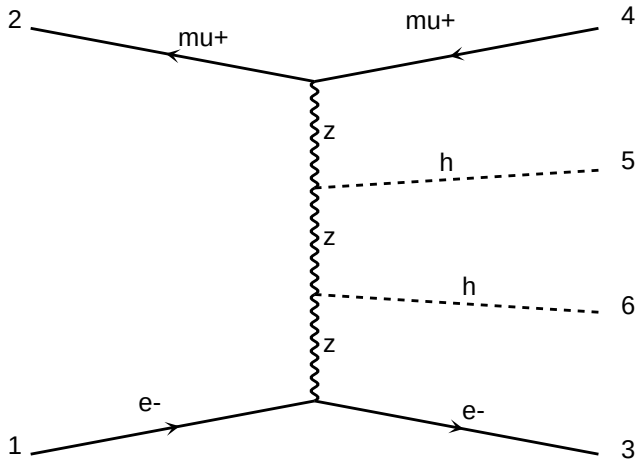

diagram 3 QCD=0, QED=4

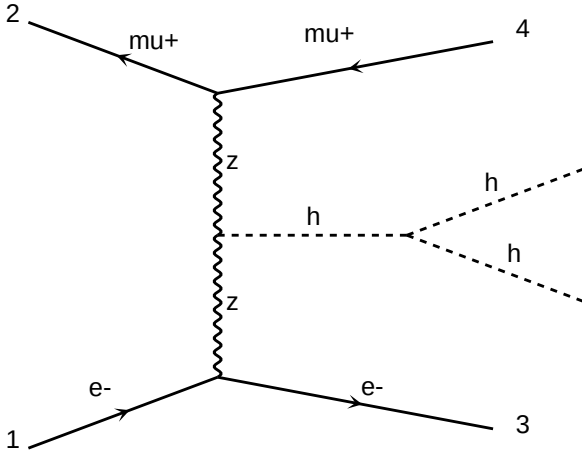

diagram 4 QCD=0, QED=4

What the numbered diagrams contain:

1 - HHZZ contact interaction (two Higgs bosons at one vertex)

2, 3 - separate Higgs emission through two HZZ couplings

4 - virtual Higgs H* -> HH (trilinear self-coupling)

Exact original MG5 diagram numbering retained; all four diagrams interfere.

Original MadGraph5_aMC@NLO diagrams

Charged current | single Higgs

Process: e- mu+ -> ve vm~ h QCD=0

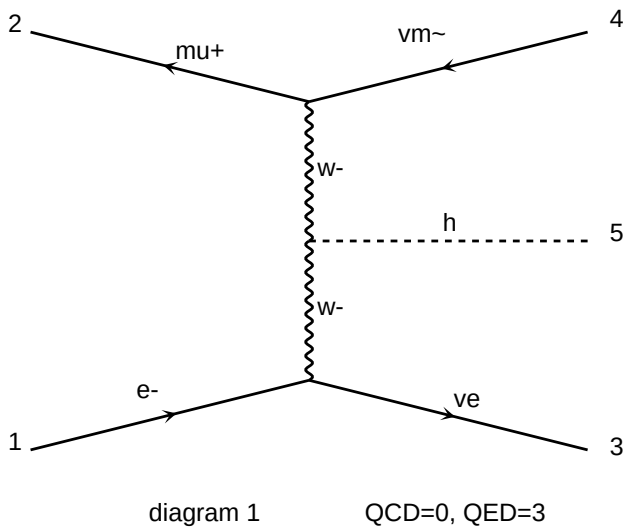

diagram 1 QCD=0, QED=3

Diagram 1: Standard Model single-Higgs production

The Higgs couples to the exchanged W bosons through the HWW vertex.

Exact original MG5 diagram numbering retained.

Original MadGraph5_aMC@NLO diagrams

Neutral current | single Higgs

Process: e- mu+ -> e- mu+ h QCD=0

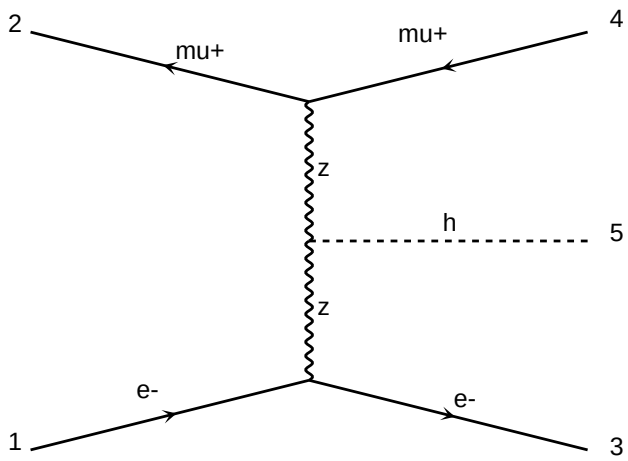

diagram 1 QCD=0, QED=3

Diagram 1: Standard Model single-Higgs production

The Higgs couples to the exchanged Z bosons through the HZZ vertex.

Exact original MG5 diagram numbering retained.